

Multiple Choice

1. **Answer:** C

Options: A) palatine processes.; B) palatine and frontonasal processes.; C) palatine, frontonasal and secondary nasal processes.; D) palatine, frontonasal, secondary nasal processes and mandibular processes.

Note: Complete bilateral cleft palate occurs when there is a failure in the fusion of the palatine processes, which form the hard palate, and the frontonasal and secondary nasal processes, which are critical for proper facial and oral structure development during embryonic growth.

2. **Answer:** B

Options: A) has been restricted to the sex chromosomes because of small family sizes; B) proceeded much more successfully as large numbers of DNA markers became available.; C) has determined that the number of linkage groups is about twice the number of chromosomes; D) has demonstrated that almost all of the DNA is involved in coding for genes

Note: The mapping of human chromosomes was significantly enhanced by the availability of numerous DNA markers, which facilitated the identification and localization of genes on the chromosomes, leading to a more successful and efficient mapping process.

3. **Answer:** B

Options: A) Do older adults still have personalities?; B) Does personality change with age?; C) Does personality decline as we grow older?; D) Does personality increase or decrease with age?

Note: The correct answer is "Does personality change with age?" because the study of aging and personality fundamentally revolves around understanding how an individual's personality traits may evolve or remain stable over the lifespan, which is a central theme in both psychological and health sciences research.

4. **Answer:** B

Options: A) is a measure of the level of heterozygotes in a population; B) is a measure of the proportion of variation that is due to genetic causes; C) is a measure of proportion of repeated DNA in an organism; D) is a measure of level of gene linkage

Note: Heritability quantifies the extent to which genetic variation contributes to observable traits within a population, indicating the proportion of phenotypic variation attributable to genetic factors.

5. **Answer:** B

Options: A) Cervix; B) Fallopian tube; C) Ovary; D) Uterus

Note: Fertilization of an ovum by a spermatozoon occurs in the Fallopian tube because this is the anatomical site where the sperm meets the egg after ovulation, facilitating the fusion of gametes.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Congenital adrenal hyperplasia leads to an enzyme deficiency in steroidogenesis, resulting in excess androgen production, which can cause the development of male secondary sexual characteristics in genetically female individuals, thus contributing to female pseudohermaphroditism.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because research indicates that older adults may continue to seek out memories even when faced with challenges, as they often value memory retrieval and may experience difficulty due to cognitive decline rather than prioritizing other tasks.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because exhaled air contains approximately 78% nitrogen and only about 16% oxygen, indicating that nitrogen is present in greater quantity than oxygen in the air we exhale.

9. **Answer:** True

Options: A) True; B) False

Note: Mutations that cause achondroplasia result in a gain-of-function in the fibroblast growth factor receptor 3 (FGFR 3), which leads to the abnormal regulation of bone growth and results in the characteristic short stature associated with the condition.

10. **Answer:** True

Options: A) True; B) False

Note: Older adults typically retain a wealth of knowledge and experience, which allows them to perform well on tests of semantic memory that assess their understanding of facts, concepts, and general knowledge.

Long Form Response

11. **Answer:** Togaviruses are a family of viruses that play a significant role in human diseases, primarily causing infections such as rubella and chikungunya. These viruses are characterized by their single-stranded RNA genomes and are primarily transmitted by arthropods. However, Yellow Fever is not classified among the diseases caused by togaviruses because it is caused by the flavivirus, specifically the Yellow Fever virus. Flaviviruses, which include other notable pathogens such as dengue and West Nile virus, are distinct from togaviruses in their genetic structure and classification. This distinction underscores the diverse nature of arboviruses and their varying impacts on human health.

Note: correct because it accurately identifies togaviruses as a distinct family responsible

for specific human diseases, such as rubella and chikungunya, while clarifying that Yellow Fever is caused by a different virus family, the flavivirus, thereby justifying its exclusion from diseases associated with togaviruses.

12. **Answer:** Hormones play a critical role in the aging process by regulating various physiological functions, including metabolism, stress response, and reproductive health. Key hormones such as estrogen, testosterone, and cortisol can influence cellular senescence and overall longevity. However, viaseterol, a compound thought to have potential anti-aging properties, has not been widely recognized as a viable agent against senescence due to insufficient research and clinical evidence supporting its efficacy in modulating the aging process. Additionally, the complexity of hormonal interactions in aging means that new agents like viaseterol must undergo rigorous validation to demonstrate their effects on healthspan and lifespan. As a result, the lack of comprehensive studies limits its acceptance in the scientific community.

Note: correct because it identifies the significant impact of hormones on aging and clarifies that viaseterol lacks sufficient research and clinical evidence to be recognized as an effective anti-senescence agent.